

1. Project Title

Splunk SIEM Home Lab: Windows Log Monitoring & Threat Detection

2. Introduction

Security Information and Event Management (SIEM) solutions help organizations collect, monitor, analyze, and investigate logs generated from systems and applications.

This project demonstrates the implementation of a basic SIEM lab using Splunk Enterprise and Splunk Universal Forwarder for centralized log collection and monitoring.

The project focuses on:

- Collecting Windows Event Logs
 - Forwarding logs to Splunk Enterprise
 - Monitoring Security and Application logs
 - Detecting failed login attempts using Splunk SPL queries
 - Understanding basic SOC (Security Operations Center) workflows
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3. Objectives

The main objectives of this project are:

- To understand SIEM architecture
 - To install and configure Splunk Enterprise
 - To configure Splunk Universal Forwarder on Windows
 - To collect Windows Security and Application logs
 - To analyze logs using Splunk Search Processing Language (SPL)
 - To detect suspicious login activities
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4. Technologies Used

Component	Technology
SIEM Platform	Splunk Enterprise 10.2.2
Log Forwarding	Splunk Universal Forwarder
Host OS	Linux

Component	Technology
Client Machine	Windows 10
Virtualization	Oracle VirtualBox
Log Type	Windows Event Logs
Query Language	SPL (Search Processing Language)

5. System Architecture

Architecture Flow

Windows Machine > Splunk Universal Forwarder > Port 9997 > Splunk Enterprise Server (Linux) > Search & Detection

6. Project Setup

6.1 Splunk Enterprise Installation

Steps performed:

1. Downloaded Splunk Enterprise for Linux
2. Installed Splunk on Linux VM
3. Started Splunk service
4. Created administrator credentials
5. Accessed Splunk Web Interface on port 8000

Command used:

```
sudo ./splunk start --accept-license --run-as-root
```

6.2 Configuring Receiving Port

To receive logs from the Windows machine:

1. Opened Splunk Web
2. Navigated to: Settings → Forwarding and Receiving
3. Enabled receiving

4. Configured port:

9997

6.3 Installing Splunk Universal Forwarder

Steps performed on Windows:

1. Downloaded Splunk Universal Forwarder
 2. Installed the forwarder
 3. Selected Windows Event Logs:
 4. Security Logs
 5. Application Logs
 6. System Logs
 7. Configured receiving indexer IP
 8. Configured port 9997
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7. Log Collection

The following Windows logs were collected successfully:

Log Type	Description
Security Logs	Login activity and security auditing
Application Logs	Application events and warnings
System Logs	Operating system events

8. Detection Use Cases

8.1 Failed Login Detection

Objective

Detect failed login attempts that may indicate brute-force attacks or unauthorized access attempts.

Event Code

4625

SPL Query

```
index=* sourcetype=WinEventLog:Security EventCode=4625
```

Description

This query searches Windows Security logs for failed login events.

Result

The failed login attempts were successfully detected and displayed in Splunk.

8.2 Successful Login Detection

Event Code

4624

SPL Query

```
index=* sourcetype=WinEventLog:Security EventCode=4624
```

Purpose

Used to monitor successful authentication events.

8.3 Privilege Assignment Detection

Event Code

4672

SPL Query

```
index=* sourcetype=WinEventLog:Security EventCode=4672
```

Purpose

Detects special privilege assignments to new logon sessions.

9. Attack Simulation

To generate failed login events:

1. Locked the Windows machine
2. Entered incorrect password multiple times
3. Generated EventCode 4625 logs
4. Verified detection in Splunk

This simulated a basic brute-force or unauthorized login attempt.

10. Sample Log Analysis

Important fields analyzed:

Field	Description
EventCode	Type of security event
Account_Name	Username involved
ComputerName	Source system
Keywords	Audit Success or Failure
TaskCategory	Type of event activity
SourceName	Source generating the event

11. Challenges Faced

During implementation, the following challenges were encountered:

- VirtualBox kernel driver issue on Linux

- Initial log forwarding configuration issues
- Security logs not appearing immediately
- Understanding Windows Event IDs
- SPL query syntax errors

These issues were resolved through configuration corrections and troubleshooting.

12. Learning Outcomes

This project helped in understanding:

- SIEM architecture and workflow
 - Centralized log monitoring
 - Windows Event Logging
 - Splunk deployment basics
 - SPL query writing
 - Security event analysis
 - Threat detection concepts
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13. Future Enhancements

Future improvements for the project:

- Create dashboards for login monitoring
 - Configure real-time alerts
 - Add Linux log monitoring
 - Integrate Sysmon logs
 - Implement brute-force detection rules
 - Add visualization panels
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14. Conclusion

This project successfully implemented a basic SIEM environment using Splunk Enterprise and Splunk Universal Forwarder.

Windows Security and Application logs were collected, forwarded, and analyzed successfully. Failed login attempts were simulated and detected using SPL queries.

The project provided practical exposure to SIEM operations, centralized logging, and security event monitoring similar to real-world SOC environments.

15. References

- Splunk Official Documentation
 - Microsoft Windows Event ID Documentation
 - Splunk SPL Documentation
 - Oracle VirtualBox Documentation
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Project: Splunk SIEM Home Lab: Windows Log Monitoring & Threat Detection

Domain: Cybersecurity / SIEM / SOC Operations